Journal of Neurology (2026) 273:81
https://doi.org/10.1007/s00415-025-13598-9

ORIGINAL COMMUNICATION

Is Parkinson's disease getting older? Demographic and modifiable determinants of age at onset in a large Italian tertiary care cohort

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Received: 18 November 2025 / Revised: 20 December 2025 / Accepted: 23 December 2025 / Published online: 16 January 2026
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Abstract

Background Parkinson's disease (PD) is a heterogeneous neurodegenerative disorder, with age at onset (AAO) influenced by genetic and environmental factors. This study aimed to (i) examine temporal trends in PD AAO over the past four decades in a large tertiary care cohort; and (ii) identify independent associations between AAO and a comprehensive set of demographics, cardiovascular/dysmetabolic, lifestyle, and environmental variables.

Methods We conducted a retrospective observational study of 13,220 individuals diagnosed with PD between 1998 and 2024 at a tertiary referral center in Italy, excluding genetically confirmed cases. Demographic, clinical, and exposure data were collected via questionnaires and interviews. Temporal trends in AAO were analysed in relation to national population ageing. Independent predictors of AAO were identified using multivariable linear regression with cross-validation and multiple imputation. Age- and sex-stratified analyses were also performed.

Results AAO increased over time in parallel with demographic ageing ($R^2 = 0.94$), without evidence of additional temporal acceleration ($P = 0.852$). An earlier AAO was independently associated with a positive family history of PD ($\beta = -2.96$, $P < 0.0001$), pollutant exposure ($\beta = -3.50$, $P < 0.0001$), and higher education level ($\beta = -0.52$, $P < 0.0001$). In contrast, type 2 diabetes mellitus ($\beta = +2.63$, $P < 0.0001$) and hypertension ($\beta = +2.24$, $P < 0.0001$) were associated with later onset. Among women, later menarche was linked to delayed AAO ($\beta = +0.60$, $P < 0.0001$).

Conclusions While demographic ageing largely accounts for the temporal increase in AAO, modifiable exposures, particularly cardiovascular/dysmetabolic comorbidities and environmental toxins, can significantly influence disease timing, though causal interpretation is limited by accumulation bias.

Keywords Parkinson's disease · Age at onset · Demographic aging · Epidemiology · Modifiable risk factors

Introduction

Parkinson's disease (PD) is a frequent, progressive neurodegenerative disorder characterized by motor symptoms including bradykinesia, rigidity, and resting tremor, as well as a wide array of non-motor manifestations [1]. From 1990 to 2015, the number of people with PD doubled to over 6 million. Driven principally by ageing, this number is projected to double again to over 12 million by 2040, a trend often referred to as the "Parkinson pandemic" [2]. In addition to ageing, environmental and lifestyle factors have been implicated in this rising incidence, either independently or through synergistic effects [2]. Nonetheless, epidemiological studies examining PD incidence have produced heterogeneous results, with some high-income countries reporting increasing incidence rates [3], whereas other studies have observed divergent patterns [4]. These discrepancies likely reflect variations in healthcare access, diagnostic criteria, and socio-environmental contexts across different populations.

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Although advancing age remains the strongest risk factor for PD, the age at onset (AAO) may vary significantly among individuals. Genetic contributions, particularly polygenic risk scores (PRS) derived from genome-wide association studies, explain part of this variability, but growing evidence suggests that environmental exposures and lifestyle factors substantially influence AAO [5].

Several modifiable lifestyle factors, including smoking, caffeine intake [6], sociodemographic characteristics [7], environmental toxicants [8], and cardiovascular/dysmetabolic comorbidities [9], have been implicated in PD risk and AAO. Reproductive and sex-specific variables, particularly age at menarche and menopause, have also been associated with variations in disease expression among women [10].

Despite this growing body of literature, integrative models assessing the combined influence of these factors on the timing of PD onset remain limited. In this study, we aim to: i) characterize temporal trends in PD AAO over the past four decades in our tertiary outpatient clinic cohort; ii) identify independent associations between AAO and a comprehensive set of demographic, cardiovascular/dysmetabolic, lifestyle, and environmental variables. By exploring these contributors within a unified framework, we focus on elucidating the modulators of disease timing and identify potential targets for early intervention and prevention.

Materials and methods

Population selection

We conducted a retrospective observational study including patients with a diagnosis of PD according to the UK Brain Bank criteria [11] and referring to the Parkinson Institute of Milan from January 1998 to December 2024. Onset years in this cohort ranged from 1940 to 2024, as some patients were referred many years after symptom onset. For patients examined before 1998, demographic and exposure data were often incomplete because standardized electronic questionnaires were not introduced until 1998. Therefore, analyses requiring complete covariate information were restricted to cases with data collected from 1998 onward. Patients with a diagnosis of genetic PD (as determined by a combination of different technologies, ranging from allele-specific PCR to exome sequencing) were excluded from the analysis. Ascertainment of demographic, lifestyle, general medical, and disease-specific information for all patients was performed using a self-administered questionnaire, with data subsequently confirmed through direct interviews conducted by a neurologist specialized in movement disorders. AAO was defined as the age when the patient first noticed any cardinal motor symptom of PD. The following variables were recorded: birth date, gender, family history, education level, smoking status (categorized as current/former *vs* never), exposure to environmental and occupational pollutants (defined as occupational or non-occupational exposure to any pesticides, solvents, heavy metals, hydrocarbons) occurring before PD onset, and region of residence. General medical history was reviewed to document the presence of T2DM, hypertension, and dyslipidemia before PD onset. For female patients, age at menarche was also recorded. Additional data regarding current and past medications were collected but not analysed for the present study.

Statistical analysis

Continuous variables were reported as means and standard deviations or medians and interquartile ranges, while categorical variables were presented as counts and percentages. Differences between groups were assessed using the Mann–Whitney or the Kruskal–Wallis test for continuous variables, whereas the Chi-square test was employed for categorical variables. Post-hoc analysis was applied when appropriate.

First, to explore whether temporal trends in AAO paralleled demographic ageing in Italy over the period 1982–2019 (source: Italian National Institute of Statistics, ISTAT; https://demo.istat.it/), AAO was regressed on the average population age (wavg_age) using ordinary least squares regression. This model provided predicted values based on population ageing. Residuals, defined as the difference between observed and predicted AAO, were subsequently regressed on calendar year to assess whether AAO increased at a rate greater than demographic ageing. We excluded data beyond 2019 to avoid confounding from the COVID-19 pandemic, which disrupted healthcare access and may have influenced PD incidence via potential neuroinflammatory effects of SARS-CoV-2 [12].

Then, multivariable linear regression models were employed to identify independent predictors of AAO after checking for normality assumptions and homoscedasticity of residuals. Model selection was conducted through a stepwise procedure, and the reliability and consistency of the identified subset of predictors were tested with a cross-validation procedure. At each stage, the dataset was randomly divided into two halves. Predictors were selected in the first half (training set), and the resulting model was tested in the second half (testing set). The procedure was repeated 200 times with different random splits. A predictor was identified as reliable only when it was chosen and validated by the cross-validation procedure in at least 70% of cases. Sensitivity analyses were performed to confirm the stability of the models, including analyses with imputed missing data and validation across an extended sample. Missingness for lifestyle and environmental exposure ranged from 10% to nearly 50%, likely due to omissions in self-reported questionnaires and

recall bias [13]. Given the large sample size and the missing at random (MAR) assumption, we applied multiple imputation by chained equations (MICE) with 30 imputations, in line with standard guidance to ensure robust estimates. The distributions of key demographic, cardiometabolic, lifestyle, and environmental variables were highly similar between the complete-case and imputed datasets (Supplementary Table S3), with standardized mean differences uniformly < 0.01. Additionally, sex-stratified models were developed to explore potential sex-specific determinants of age at PD onset. For the female subgroup, age at menarche was included as an additional covariate.

All statistical tests were two-tailed, with a significance threshold set at $P < 0.05$. Analyses were conducted using SPSS version 25 (IBM Corporation, Armonk, USA), Jamovi version 2.3.28 (https://www.jamovi.org), and SAS version 9.4 (SAS Institute, Cary, NC).

Results

Association between population ageing and PD onset

After excluding 238 genetically confirmed PD cases, the final cohort available for analysis consisted of 13,220 patients with PD. We evaluated whether trends in age at PD onset mirrored the ageing of the Italian population. The model revealed a strong positive association between wavg_age and AAO ($R^2 = 0.94$), indicating that demographic ageing accounted for the majority of the temporal increase in AAO between 1982 and 2019 (see Fig. 1). Residuals from the model were then regressed on calendar year. This second regression showed no significant effect ($P = 0.852$), providing no statistical evidence that AAO increased beyond what would be expected from demographic changes.

Predictors of age at onset

Comparative analyses revealed that AAO was significantly lower among patients with a positive family history of PD (median 60 years, IQR = 52–68 vs. 63 years, IQR = 55–70; $P < 0.0001$), males (median 62 years, IQR = 54–69 vs. 63 years, IQR = 55–70; $P < 0.0001$), smokers (median 62 years, IQR = 54–69 vs. 63 years, IQR = 55–70; $P < 0.0001$), and those exposed to pollutants (median 60 years, IQR = 52–67 vs. 64 years, IQR = 56–71; $P < 0.0001$), particularly industrial solvents (35.1%). Occupational and environmental exposures among the PD cohort are reported in Supplementary Table S1. Conversely, AAO was significantly higher in individuals reporting coffee consumption (median 63 years, IQR = 48–70; vs. 62 years, IQR = 47–69; $P = 0.018$) and in those with pre-existing T2DM (median 68 years, IQR = 61–71 vs. 63 years, IQR = 55–69; $P < 0.0001$), hypertension (median 67 years, IQR = 60–72 vs. 62 years, IQR = 54–69; $P < 0.0001$), and dyslipidemia (median 65 years, IQR = 59–70 vs. 63 years, IQR = 55–70; $P = 0.002$), see Fig. 2. Counts and percentages for each variable are reported in Supplementary Table S2. Spearman correlation analysis showed significant inverse relationships between AAO and years of education ($r = -0.159$, $P < 0.0001$) and direct correlations between AAO and duration of pollutant exposure ($r = 0.159$, $P < 0.0001$), as well as age at menarche in females ($r = 0.102$, $P < 0.0001$).

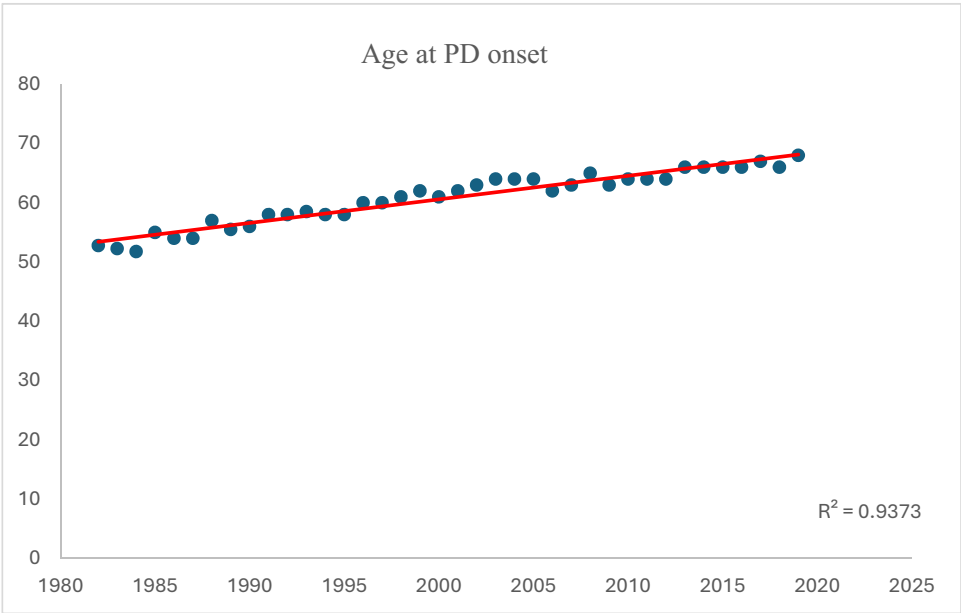

Fig. 1 Observed annual mean AAO values (blue dots) are plotted against calendar years (from 1982 to 2019). The red line represents predicted AAO values obtained from a linear regression model in which AAO was regressed on the national average population age

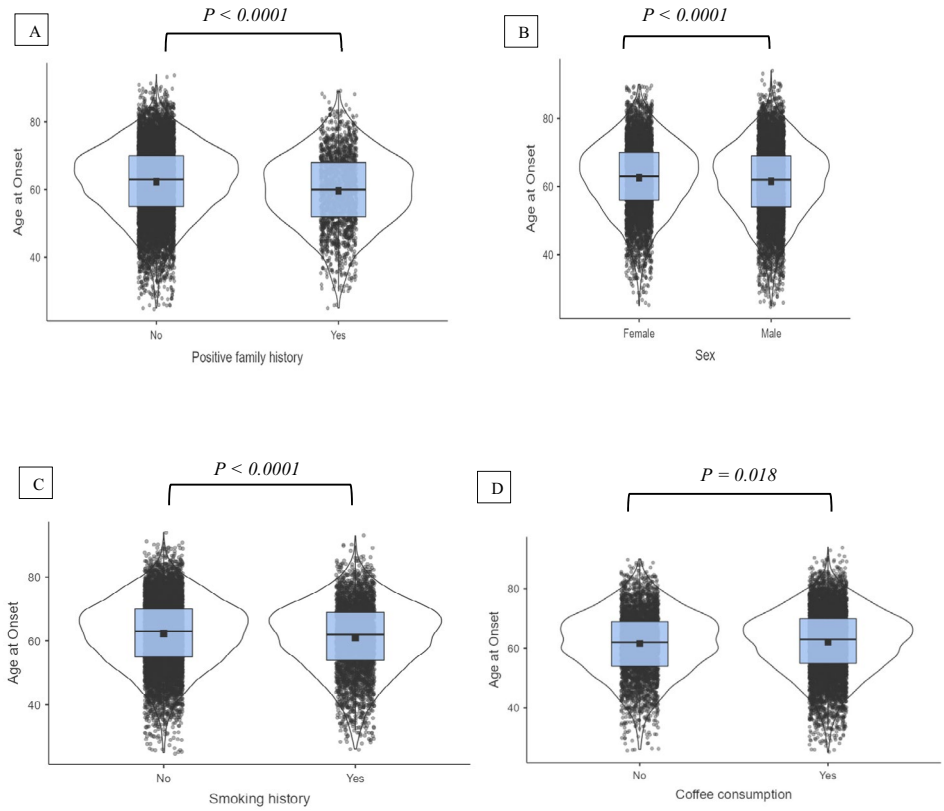

Fig. 2 Box and violin plots show the comparison of median age at PD onset after stratification by demographic **A**, **B**, lifestyle **C**, **D**, environmental **E**, and cardiovascular/dysmetabolic **F**–**H** factors using the Mann- Whitney test

Multivariable regression identified five independent predictors of AAO: T2DM (β = 3.91; $P < 0.0001$), hypertension (β = 4.39; $P < 0.0001$), family history of PD (β = − 2.72; $P < 0.0001$), exposure to environmental/occupational pollutants (β = − 4.12; $P < 0.0001$), and education level (β = − 0.43; $P < 0.0001$). The ageing index (wavg_age), as a covariate, did not attenuate these associations. In the age-adjusted model, all previously identified predictors remained significant, with wavg_age also emerging as a strong positive predictor of AAO (β = 2.23; $P < 0.0001$); see Table 1 for adjusted β and *P*-values.

Sensitivity analyses using models with imputed missing data confirmed the stability of these results (see Table 2). In particular, T2DM and hypertension were validated as the strongest factors associated with later AAO (β = 2.63, $P < 0.0001$; β = 2.24, $P < 0.0001$), whereas exposure to pollutants was the strongest anticipatory factor (β = −3.50, $P < 0.0001$).

Age-stratified analysis

Patients were stratified by age at PD onset into five groups: 20–45 years, 46–60 years, 61–70 years, 71–80 years, and 80 years or older. Mean years of education were significantly lower in patients with later onset, decreasing from 12 ± 4 years in the youngest group to 9 ± 4 years in the oldest ($P < 0.0001$). The prevalence of T2DM, hypertension, and dyslipidemia increased significantly with higher AAO ($P < 0.0001$ for all). In contrast, the proportions of patients with a positive family history of PD and those with environmental or occupational pollutant exposure declined across older onset groups ($P < 0.0001$). Similarly, smoking prevalence decreased with older AAO ($P < 0.0001$). Caffeine consumption remained high across all groups and did not differ significantly ($P = 0.0716$). Descriptive statistics (means ± standard deviations or counts/percentages) and post-hoc *P*-values across AAO strata are reported in Tables 3 and 4.

Sex-stratified analysis

Among women, stepwise regression identified the same core predictors as in the overall cohort: T2DM (β = 2.87; $P = 0.0063$), hypertension (β = 4.87; $P < 0.0001$), family history of PD (β = −2.61; $P < 0.0001$), pollutant exposure (β = − 3.70; $P < 0.0001$), and education (β = −0.53; $P < 0.0001$). In addition, age at menarche emerged as a significant independent predictor (β = 0.60; $P < 0.0001$),

Fig. 2 (continued)

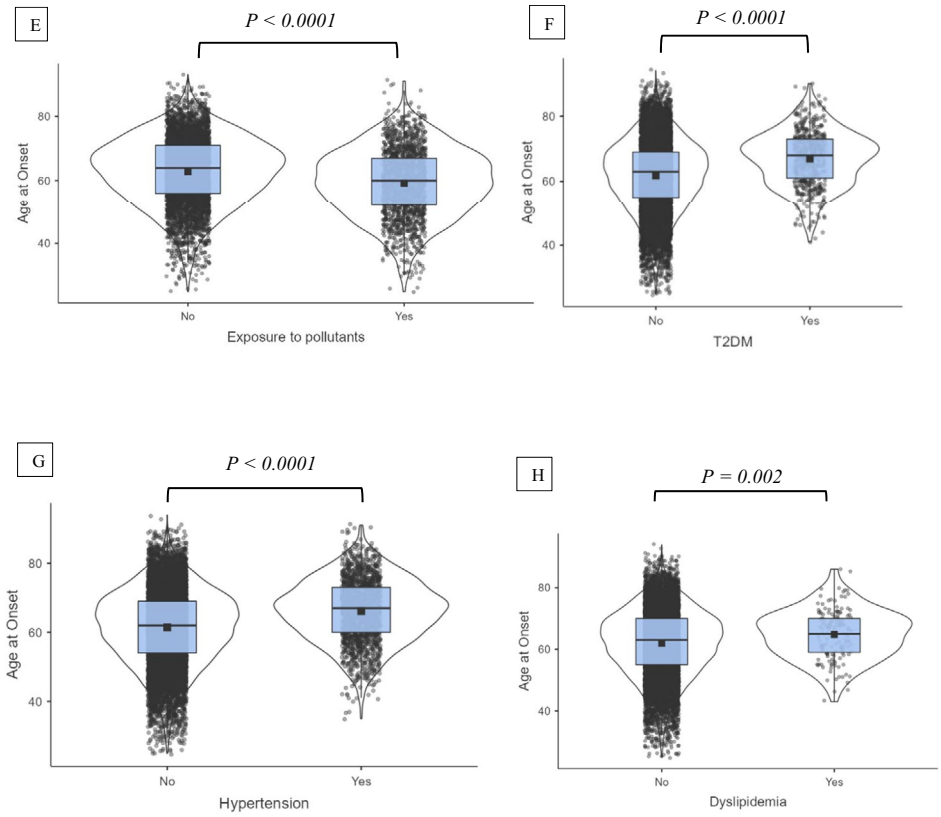

Table 1 Identification of significant predictors of age at PD onset using regression analysis with the stepwise selection method. Cross-validation was employed to assess internal validity

Variables	Estimate	Standard Error	P-value
Intercept	− 25.99	3.33	<.0001
T2DM	2.83	0.56	<.0001
Exposure to pollutants	− 3.03	0.28	<.0001
Positive family history	− 2.37	0.31	<.0001
Hypertension	2.93	0.32	<.0001
Education level	− 0.57	0.03	<.0001
Wavg_age	2.22	0.08	<.0001

Table 2 Model adjustment results after implementation of missing data and sensitivity analysis

Variables	Estimate	Standard Error	P-value
Intercept	4.89	0.62	<.0001
T2DM	2.63	0.14	<.0001
Exposure to pollutants	− 3.50	0.06	<.0001
Positive family history	− 2.96	0.08	<.0001
Hypertension	2.24	0.09	<.0001
Education level	− 0.52	0.01	<.0001
Wavg_age	1.52	0.01	<.0001

indicating a potential role of hormonal exposure in modulating age at PD onset among women.

In the male subgroup, the same set of predictors remained significant: T2DM ($\beta = 4.49$; $P < 0.0001$), hypertension ($\beta = 4.00$; $P < 0.0001$), family history of PD ($\beta = -2.65$; $P < 0.0001$), pollutant exposure ($\beta = -4.16$; $P < 0.0001$), and education ($\beta = -0.35$; $P < 0.0001$). No additional sex-specific predictors were identified. Sex-stratified analysis results are reported in Table 5.

Discussion

This study elucidates the complex interplay of demographic, cardiovascular/dysmetabolic, environmental, and reproductive factors influencing the AAO of PD within a large tertiary care cohort. Our findings highlight a substantial contribution of population ageing to increasing AAO over the past four decades, consistent with global demographic transitions [2]. Importantly, we observed that several individual-level exposures were associated with variation in disease onset.

Multiple independent predictors of AAO were identified. Among these, prevalent cardiovascular and dysmetabolic comorbidities, particularly T2DM and hypertension, showed the most substantial association with later AAO. The

Table 3 Means ± standard deviations (or counts and percentages) of different variables across age at onset strata

Variables	20–45 years	46–60 years	61–70 years	71–80 years	> 80 years	P-value
Education level	12 ± 4	11 ± 4	10 ± 4	9 ± 4	9 ± 4	<.0001
T2DM	4(0.4)	121(2.6)	198(4.2)	157(6)	25(6.8)	<.0001
Exposure to pollutants	179(35)	708(29)	526(22)	228(17)	17(9)	<.0001
Positive family history	200(20.4)	793(17)	633(13)	274(11)	32(9)	<.0001
Dyslipidemia	1(0.1)	44(0.95)	70(1.5)	36(1.4)	2(0.6)	0.0011
Hypertension	19(1.9)	393(8.5)	638(14)	426(17)	76(21)	<.0001
Sex, male	631(64)	2773(60)	2699(57)	1431(56)	192(52)	<.0001
Coffee consumption	582(66)	2813(67)	2914(69)	1586(69)	221(68)	0.0716
Smoking history	331(37.5)	1291(31)	1260(30)	617(27)	76(23)	<.0001

Table 4 Post-hoc analysis of different parameters across age at onset strata

Variables	20–45 vs 46–60	20–45 vs 61–70	20–45 vs 71–80	20–45 vs > 80	46–60 vs 61–70	46–60 vs 71–80	46–60 vs > 80	61–70 vs 71–80	61–70 vs > 80	71–80 vs > 80
Education level	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	0.0013	0.0027	ns
T2DM	0.0010	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	0.0112	ns
Exposure to pollutants	0.0032	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	0.0002	<.0001	0.0243
Positive family history	0.0094	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	0.0013	0.0161	ns
Dyslipidemia	0.0236	0.0002	0.0012	ns	0.0152	ns	ns	ns	ns	ns
Hypertension	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001	0.0186
Sex, male	0.0125	<.0001	<.0001	<.0001	0.0117	0.0005	0.0053	ns	ns	ns
Coffee consumption	Ns	ns	ns	ns	ns	ns	ns	ns	ns	ns
Smoking history	0.0001	<.0001	<.0001	<.0001	ns	0.0013	0.0047	0.0132	0.0129	ns

Table 5 Sex-stratified regression analysis using the stepwise selection method. Cross-validation was employed to assess internal validity

	Regression model 1 (women, n = 5494)		
Variables	Estimate	Standard Error	P-value
Intercept	59.78	1.93	<.0001
T2DM	2.87	1.05	0.0063
Exposure to pollutants	− 3.70	0.60	<.0001
Positive family history	− 2.61	0.51	<.0001
Hypertension	4.87	0.53	<.0001
Education level	− 0.53	0.05	<.0001
Age at menarche	0.60	0.14	<.0001
	Regression model 2 (men, n = 7726)		
Variables	Estimate	Standard Error	P-value
Intercept	66.18	0.53	<.0001
T2DM	4.49	0.75	<.0001
Exposure to pollutants	− 4.16	0.37	<.0001
Positive family history	− 2.65	0.45	<.0001
Hypertension	4.00	0.45	<.0001
Education level	− 0.35	0.04	<.0001

coefficients for these conditions translate to a predicted delay in the AAO of approximately 2.6 to 2.2 years. However, the interpretation of these associations requires caution, since both conditions are strongly age-related and accumulate over the lifespan. One possible explanation is that the reported associations are influenced by time-at-risk, age-dependent accumulation, and survival bias, which cannot be fully addressed by the employed retrospective design. In this light, the magnitude of the β coefficients is most likely explained by the fact that PD patients with T2DM and hypertension are older than those without these comorbidities. This suggests that T2DM and hypertension are likely markers of survival to an older age, rather than definitive causal determinants actively postponing PD onset. On the other hand, the possibility that treatments for metabolic or cardiovascular disorders may modulate PD risk or progression remains under investigation. Previous studies suggested that metabolic dysfunction may, paradoxically, delay PD onset. While metabolic disorders such as T2DM and hypertension are typically associated with detrimental effects on the nervous system, through mechanisms including cerebrovascular damage, small vessel disease, and chronic hyperglycemia-induced neurotoxicity [14], several clinical studies show a later PD onset among patients with these conditions. This apparent discrepancy likely reflects a complex interplay between the direct harmful effects of these diseases and the potential neuroprotective benefits of pharmacological treatments commonly used to manage them.

Supporting this, a previous study observed significantly higher age at PD onset of diabetic patients (66.2 ± 9.0 years) compared to non-diabetic subjects (59.6 ± 11.0 years; $P < 0.001$) [15]. Similarly, our group found that diabetic patients on any form of antidiabetic therapy developed PD more than 6 years later than non-diabetic subjects or subjects who developed T2DM after the onset of PD [16]. Although we did not assess specific medications in the present analysis, growing literature supports the repurposing potential of antidiabetic drugs for neurodegenerative diseases [17–20]. Antihypertensive medications have also emerged as repurposing candidates in PD. Exposure to adrenergic blockers was reported as one of the strongest independent predictors of older AAO in the study by Malatt et al. [21]. More recently, the use of angiotensin II receptor blockers with blood–brain barrier-penetrating properties and a high cumulative duration has been linked to a significantly reduced risk of PD [22]. However, this protective effect was not confirmed in Mendelian randomization studies [23]. Regarding dyslipidemia, its association with AAO was not retained in our final model. Nevertheless, several observational studies have reported a protective role of long-term statin use [24], although contradictory evidence has also been reported [25].

Self-reported exposure to environmental pollutants, including pesticides, solvents, and metals, was robustly associated with an earlier onset in our study. These results confirm mechanistic studies implicating environmental toxins in dopaminergic neurodegeneration through oxidative stress, microglial activation, and neuroinflammation [26]. A recent meta-analysis confirmed that toxic occupational exposures are significant risk factors for PD [27], and industrial toxins showed the highest odds ratio [28], in line with our results. Interestingly, while the prevalence of exposure decreased in patients with later AAO, Spearman correlation analysis demonstrated a direct correlation between AAO and duration of exposure among those exposed. In this regard, it is conceivable that longer exposure durations tend to be observed in patients with later onset, likely reflecting age-related accumulation rather than a causal effect on the timing of onset. Taken together, these findings reinforce the need for preventive strategies aimed at reducing environmental toxin exposure, particularly through enhanced environmental regulations, comprehensive screening for neurotoxic potential, and systematic public health approaches.

Lifestyle factors also influenced AAO. While smoking has consistently been associated with reduced PD risk in epidemiological studies, its relationship with AAO is less clear. Whether smoking influences PD AAO is still under debate: whereas some studies showed a delay in AAO [29], others reported an opposite effect [30] or no association [31]. In our cohort, ever smokers had a slightly earlier onset compared to never smokers, but the effect size was modest and not consistently significant across models. Furthermore, our study did not compare consistent non-smokers vs. consistent smokers, and smoking dosage or duration was not estimated. In contrast, patients who reported habitual coffee intake tended to experience a later onset of PD, aligning with previous findings suggesting a neuroprotective role of caffeine [5]. However, as with smoking, the lack of detailed data on dosage and duration limits the strength of causal inference.

An unexpected finding was the association between higher educational attainment and earlier AAO, which was retained as a significant predictor in cross-validated regression models. While a recent Mendelian randomisation study has identified positive associations between educational attainment and PD risk [7], the direction and interpretation of this relationship remain uncertain. Frigerio et al. also observed that subjects with higher education had an increased risk of PD [32]. Several explanations could account for this association, including increased health surveillance among more educated individuals, unmeasured socioeconomic exposures, or differing lifetime patterns of physical activity [33]. In the context of our findings, a plausible explanation is ascertainment bias inherent to tertiary referral centers. Although all Italian citizens have access to specialized care resources through the National Health Service, high educational attainment is strongly associated with greater health literacy and proactive healthcare-seeking

behaviour. Therefore, it is conceivable that highly educated patients may recognize subtle motor symptoms of PD and are more likely to seek immediate specialist consultation. Another hypothesis is that personality traits may contribute as well. Indeed, PD patients are known to present a personality profile of low novelty seeking and high harm avoidance [34]. A study on lifelong learning identified that novelty-seeking personalities are linked to more engagement in new educational activities and occupational changes, but not to a sustained increase in engagement over time [35]. This could imply that non-novelty-seeking individuals are more consistent in pursuing and completing long-term educational goals, thereby achieving higher educational attainment. While this remains speculative, it offers one possible hypothesis that could help explain our findings. However, further research is needed to thoroughly assess the relationship between educational attainment and personality traits and how this may influence AAO.

From a sex-specific perspective, age at menarche emerged as an independent predictor of AAO among women, with later menarche associated with delayed PD onset. The role of estrogens in dopaminergic pathway degeneration remains inconclusive, as existing studies have yielded mixed results, including beneficial, detrimental, and null effects [36]. Our results replicate previous findings [37] but are in contrast with other studies reporting no association between age at menarche and PD [38]. A possible explanation for this discrepancy is that we did not assess other reproductive-related characteristics (i.e., age at menopause, regularity of menstrual periods, number of pregnancies, duration of estrogen exposure, use of contraceptives or hormonal replacement therapy); alternatively, early puberty has been linked to accelerated epigenetic ageing, which may increase the susceptibility to PD development [39].

Overall, our findings highlight the complex and multifactorial determinants of AAO in PD. Although demographic ageing strongly accounts for the upward temporal trend in AAO, our findings highlight that individual-level exposures, particularly cardiovascular/dysmetabolic comorbidities and environmental pollutants, exert independent and substantial effects on the timing of disease onset. These results provide a strong rationale for future studies to examine whether reducing environmental toxin exposure and improving metabolic health may delay PD onset.

While our study offers one of the largest single-center datasets examining AAO in PD, several limitations should be acknowledged. First, the retrospective design and reliance on self-reported exposures introduce potential for recall bias. Although the design inherently limits causal inference, it should be noted that this approach represents the only methodologically feasible option for examining risk factors that precede disease onset by several decades, as prospective studies would require follow-up periods that exceed the practical constraints of clinical research. Second, the absence of direct pharmacological, imaging, or biomarker data prevented deeper mechanistic insights. Nevertheless, the comprehensive clinical and exposure data collected allowed for robust epidemiological associations, providing solid groundwork for future studies incorporating biomarker-driven investigations. Third, although we lacked an external validation cohort, which limits the generalizability of our results, the large cohort size and use of robust statistical methods helped mitigate potential biases and strengthened the validity of our findings.

The observed associations highlight the complex interplay of demographic, environmental, cardiovascular/metabolic, lifestyle, and sex-specific factors influencing AAO in PD. While population ageing appears to be the dominant driver of temporal changes in onset, individual-level exposures may contribute to heterogeneity in AAO. In particular, environmental exposures emerge as potentially relevant factors from a public health perspective, whereas associations involving cardiovascular/metabolic comorbidities may reflect different explanations, including age-related processes and other biological mechanisms. Future longitudinal studies will be required to determine whether modifying these exposures may influence disease onset.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00415-025-13598-9.

Acknowledgements The authors thank all patients and families for their contribution, as well as the Associazione Italiana Parkinsoniani and the Fondazione Pezzoli per la Malattia di Parkinson (Milan, Italy) for the long-lasting support.

Author contributions Conception and design of the study: GP. Drafting of the manuscript and tables: EC, AB. Statistical analysis: EC, AB. Acquisition and analysis of data: EC, GS, MZ, FDS, PS, AZ, GP. Interpretation of data: EC, DC, IUI, GP. Critical revision of the article: DC, GS, MZ, FDS, AZ, PS, IUI, GP. All authors discussed the results and contributed to the final manuscript.

Funding This work was supported by the unconditional and independent contribution of the 'Fondazione Pezzoli per la Malattia di Parkinson', Milan (Italy), (Italian '5 × 1000' funding), a Research Foundation linked to the 'Associazione Italiana Parkinsoniani' (AIP).

Availability of data and materials The dataset used during the current study is available from the corresponding author upon reasonable request.

Declarations

Conflicts of interest IUI received lecture honoraria and research funding from Medtronic Inc. and Boston Scientific. IUI is a Newronika S.p.A. consultant and shareholder and received funding for research activities from Newronika S.p.A. The other authors declare that they have no competing interests.

Ethical approval and consent to participate Study procedures were approved by the local Ethics Committee (Comitato Etico Territoriale Lombardia 3, protocol 3499/23), and this research was conducted in

accordance with the principles of the Declaration of Helsinki and its subsequent amendments. All participants gave their written informed consent.

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